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Extrapolative Bubbles and Trading Volume

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1 Big picture

- **Question.** Why do trading volume and prices both explode during bubbles? Standard extrapolative-belief models can generate price run-ups, but they have difficulty generating very high volume because similarly optimistic extrapolators should mostly buy and hold.
- **Setting.** The paper studies the 2014–2015 Chinese stock market bubble using account-level transaction data from one of the largest Chinese brokerage firms. The episode has the classic bubble pattern: initial good news, a long run-up, very high trading volume, and a sharp crash.
- **Main idea.** The paper combines **extrapolative beliefs** with the **disposition effect**. Extrapolation makes investors want to buy stocks with strong past returns. The disposition effect makes investors want to sell stocks once those same strong returns generate paper gains.
- **Core mechanism.** A disposition extrapolator is quick to buy a rising stock, but also quick to sell it after the rise continues and creates a gain. The investor therefore cycles through positions rather than simply holding. This “belief–preference tension” produces high volume.
- **Model contribution.** The model provides a micro-foundation for the “flip-flopping” behavior needed to generate bubble volume. Investors alternate between cash and stocks because extrapolative beliefs push them into winners, while realization utility pushes them to lock in gains.
- **Empirical design.** The paper measures each investor’s extrapolation and disposition *before* the bubble, using transactions from 2005–2013. It then asks whether investors with high prebubble extrapolation and high prebubble disposition trade more during the 2014–2015 bubble.
- **Main market-level result.** At the bubble peak, disposition extrapolators increase monthly trading volume by almost **800%**. Pure extrapolators increase volume by about **500%**. The gap shows that extrapolation alone is not enough; the disposition effect matters for volume.
- **Main investor-level result.** A one-standard-deviation increase in extrapolation is associated with a **402%** increase in peak volume in the baseline account regression; a one-standard-deviation increase in disposition is associated with a **460%** increase. In the full specification, both remain economically large and statistically significant.
- **Main stock-level result.** Stocks traded more by extrapolators and disposition-prone sellers have higher weekly turnover. In the baseline stock-level regression, a one-standard-deviation increase in stock-level extrapolation raises weekly turnover by **0.04**; a one-standard-deviation increase in stock-level disposition raises it by **0.02**.
- **Magnitude result.** A structural counterfactual estimates $\theta = 0.64$ for the extrapolation horizon and $\beta = 0.54$ for the disposition force. Removing disposition extrapolators lowers model-implied peak volume from **0.46** to **0.36**; the mechanism therefore increases peak volume by roughly **30%**.
- **Price result.** The paper also finds evidence consistent with extrapolators pushing prices up in the run-up and down in the crash. A one-standard-deviation increase in stock-level extrapolation predicts about **35 basis points** higher next-week returns during the run-up, and the IV estimate is about **70 basis points**.

- **Identification caveat.** The volume evidence is the paper's cleanest empirical evidence because investor types are measured before the bubble and then used to explain subsequent trading. The price evidence is suggestive rather than fully causal, though predictive and IV regressions reduce the reverse-causality concern.
- **Economic takeaway.** Bubbles can be fueled by the joint effect of nonstandard beliefs and nonstandard preferences. Extrapolation gets investors into rising assets; realization utility gets them out after gains; together they generate both price pressure and excessive trading.

2 Data and measurement (market setting and key objects)

2.1 Why the Chinese bubble is a useful laboratory

- The 2014–2015 Chinese stock market episode displays the empirical symptoms of a bubble: rapid price appreciation, heavy public attention, widespread discussion of overvaluation, high trading volume, and a sharp reversal.
- The Shenzhen Component Index increased from about **8,332** to **18,098** during the run-up and then fell by almost **40%** in one month after the mid-June 2015 crash began.
- Trading volume rose even more dramatically than prices, reaching about four times its prebubble level.
- Retail investors are especially important in China. Individual investors hold a large share of tradable shares and account for most trading volume, making account-level data especially informative.

Interpretation. The setting is useful because the bubble is large, recent, and retail-investor driven. This lets the paper connect a behavioral theory directly to observed account-level trading.

2.2 Bubble timeline

- **Prebubble period:** roughly January 2014 to November 2014.
- **Run-up:** roughly December 2014 to May 2015; the figure-level dating is November 18, 2014 to June 12, 2015.
- **Crash:** roughly June 2015 to August/September 2015; the figure-level dating is June 13, 2015 to September 15, 2015.

Interpretation. The paper uses the prebubble period to establish a quiet baseline, the run-up to test high-volume predictions, and the crash to run placebo-style comparisons.

2.3 Account-level transaction data

- The data come from one of the largest brokerage firms in China and contain complete transaction records for exchange-traded assets.
- Before filters, the brokerage data cover almost **three million** accounts, about **5%** of the investor population.
- After dropping inactive accounts, the sample is reduced to about **1.2 million** accounts.
- The main sample focuses on individual investors with accounts opened before 2014, nontrivial but relatively small balances, and enough prior transactions to estimate behavioral measures.
- The final main sample contains about **583,859** investors with transactions from 2005 to 2016.

Interpretation. The paper intentionally measures behavioral traits before the bubble begins. This prevents the key explanatory variables from being mechanically created by the bubble itself.

2.4 Investor-level characteristics

- **Account balance:** average RMB holdings, measured in millions.
- **Experience:** years since account opening.
- **Trading counts:** number of buys and sells.
- **Turnover:** total trading volume divided by average account balance.
- **Returns:** average monthly RMB return divided by average holdings.
- **Demographics:** gender, age, education, and survey-based information where available.
- **Special-account indicators:** margin-account ownership and prior trading in warrants or structured funds.

Interpretation. These variables help distinguish the paper’s mechanism from alternative explanations such as leverage, experience, gambling preference, risk tolerance, sophistication, and prior bubble trading.

2.5 Measuring extrapolation: DOX

The paper measures investor i ’s degree of extrapolation from initial buying behavior before the bubble:

$$\text{DOX}_i = \frac{\sum_t \text{Buy}_{i,t} \text{PastRet}_t}{\sum_t \text{Buy}_{i,t}}. \quad (1)$$

- $\text{Buy}_{i,t}$ is the transaction value for investor i ’s initial buy transaction t .
- PastRet_t is the past return of the purchased stock before transaction t .
- Initial buys are purchases of stocks not currently in the investor’s portfolio.

- The main measure uses past one-month returns; a weekly-return version is also reported.

Interpretation. A high DOX_i means the investor tends to initiate positions in stocks that have recently gone up. This is interpreted as extrapolative buying or positive-feedback trading.

2.6 Measuring disposition: DOD

The disposition effect is measured using the classic realized-gains and realized-losses approach:

$$PGR_i = \frac{\# \text{ Realized Gains}_i}{\# \text{ Realized Gains}_i + \# \text{ Paper Gains}_i}, \quad (2)$$

$$PLR_i = \frac{\# \text{ Realized Losses}_i}{\# \text{ Realized Losses}_i + \# \text{ Paper Losses}_i}. \quad (3)$$

The paper reports both a difference measure and a ratio measure:

$$DOD_i^D = PGR_i - PLR_i, \quad (4)$$

$$DOD_i^R = \frac{PGR_i}{PLR_i}. \quad (5)$$

- A high PGR_i means the investor often sells winners.
- A low PLR_i means the investor is reluctant to realize losses.
- A high DOD_i therefore indicates stronger disposition behavior.

Interpretation. Disposition-prone investors are more likely to sell stocks after gains. In a bubble, this creates exactly the selling pressure needed for high turnover among investors who are also extrapolative buyers.

2.7 Investor groups

- **Disposition extrapolators:** investors with both DOX and DOD above the median.
- **Pure extrapolators:** investors with DOX above the median but DOD below the median.
- **Others:** all remaining investors, including many investors who are disposition-prone but not extrapolative.

Interpretation. The key comparison is between disposition extrapolators and pure extrapolators. Both buy winners, but only the former are also eager to sell winners.

2.8 Stock-level exposure measures

For each stock j and week t , the paper constructs stock-level exposure to extrapolative buying:

$$DOX_{j,t} = \sum_i \left(\frac{Buy_{i,j,t}}{\sum_i Buy_{i,j,t}} \right) DOX_i. \quad (6)$$

It similarly constructs stock-level exposure to disposition-prone selling:

$$\text{DOD}_{j,t} = \sum_i \left(\frac{\text{Sell}_{i,j,t}}{\sum_i \text{Sell}_{i,j,t}} \right) \text{DOD}_i. \quad (7)$$

Interpretation. A stock has high $\text{DOX}_{j,t}$ if it is being bought by extrapolators in that week. It has high $\text{DOD}_{j,t}$ if it is being sold by investors with strong disposition effects.

2.9 Descriptive facts

- The median account balance is about **130,000 RMB**; the average balance is about **220,000 RMB**.
- The median investor has more than seven years of account experience and makes many trades before the bubble.
- The ownership of margin accounts is low in the main sample, around **2%**.
- Chinese investors in the sample are generally extrapolative: the 25th percentile of the monthly-return-based DOX measure is positive.
- The disposition effect is also prevalent: the 75th percentile of the difference measure $\text{PGR} - \text{PLR}$ is positive, and the ratio measure PGR / PLR is well above one for many investors.
- Ownership breadth is low. The median stock is held by only about **0.07–0.08%** of investors, implying highly heterogeneous and concentrated portfolios.

Interpretation. The descriptive evidence supports the model’s assumptions: investors chase past returns, realize gains more readily than losses, and begin with different holdings.

3 Models and empirical specifications (all equations + interpretation)

3.1 Dividend process and market environment

The stock is a claim to a final dividend D_T :

$$D_T = D_0 + d_1 + \cdots + d_T, \quad (8)$$

where shocks d_t are normally distributed and independent over time. There are two assets: cash with zero return and one risky stock with fixed supply Q .

Interpretation. Fundamentals can receive positive shocks. These shocks start the run-up, but the model asks why prices and volume continue to rise after the shocks stop.

3.2 Extrapolative signal

Investors form an extrapolative signal from past price changes:

$$X_t = (1 - \theta) \sum_{k=1}^{t-1} \theta^{k-1} (P_{t-k} - P_{t-k-1}) + \theta^{t-1} X_1, \quad (9)$$

where $0 < \theta \leq 1$.

- X_t is an exponentially weighted average of past price changes.
- A lower θ means investors place more weight on recent returns and are therefore more extrapolative.
- A higher recent return raises X_t , creating optimism about future price changes.

Interpretation. Extrapolation turns past returns into current demand. Positive shocks can therefore set off a feedback loop: higher prices create more optimism, which creates further buying pressure.

3.3 Belief formation

Investor i 's expected price change from t to $t + 1$ is

$$E_{i,t} \Delta P_{t+1} = \gamma X_t + (1 - \gamma)(D_t - P_t) + \varepsilon_{i,t}. \quad (10)$$

- X_t is the extrapolative signal.
- $D_t - P_t$ is a value signal: if price is below the current fundamental benchmark, the stock appears undervalued.
- γ controls the weight on extrapolation.
- $\varepsilon_{i,t}$ is idiosyncratic belief noise.

Interpretation. Prices can rise above fundamentals when the extrapolative signal dominates the negative value signal. This is how the model generates the bubble price path.

3.4 Realization utility and the disposition effect

The investor maximizes expected next-period wealth plus utility from realizing gains:

$$E_t(W_{t+1}) + \beta(P_t - \bar{P}_t)(N_{t-1} - N_t) \mathbf{1}\{N_{t-1} > 0 \text{ and } N_{t-1} > N_t\}. \quad (11)$$

- $\bar{P}_t$ is the investor's reference price, proxied by the average purchase price.
- $P_t - \bar{P}_t$ is the paper gain on the stock position.
- N_t is the number of shares held after trading at date t .

- β measures the strength of realization utility or the disposition effect.

Interpretation. When the stock is trading at a gain, selling generates realization utility. Thus the same rising price that attracts extrapolative buyers also gives existing holders a reason to sell.

3.5 Share demand and market clearing

Let $\Phi(\cdot)$ be the standard normal CDF. Cash holders switch into the stock if their expected price change is positive. Stock holders keep the stock if expected returns dominate the utility of realizing gains. Aggregate demand is

$$H_t = \Phi\left(\frac{E_t \Delta P_{t+1}}{\sigma_\varepsilon}\right) \frac{W_{X,t-1}^C}{P_t} + \Phi\left(\frac{E_t \Delta P_{t+1} - \beta(P_t - \bar{P}_t)}{\sigma_\varepsilon}\right) Q. \quad (12)$$

Market clearing requires

$$H_t = Q. \quad (13)$$

Interpretation. The first term is demand from cash investors who are attracted by positive expected returns. The second term is demand from current stock holders after accounting for the urge to realize gains. The equilibrium price balances these two forces.

3.6 Trading volume in the model

Total trading volume is

$$V_t = \frac{1}{2} \left[\Phi\left(\frac{E_t \Delta P_{t+1}}{\sigma_\varepsilon}\right) \frac{W_{X,t-1}^C}{P_t} + \Phi\left(\frac{\beta(P_t - \bar{P}_t) - E_t \Delta P_{t+1}}{\sigma_\varepsilon}\right) Q \right]. \quad (14)$$

Interpretation. The first component is buying by cash holders; the second is selling by stock holders. Since every buy must match a sell, the expression divides by two. Volume rises when both the desire to buy winners and the desire to sell winners are strong.

3.7 Baseline simulation parameters

The baseline model uses

$$T = 100, \quad D_0 = 100, \quad X_1 = 0, \quad \theta = 0.8, \quad \beta = 1, \quad \sigma_\varepsilon = 2, \quad \gamma = 0.9, \quad Q = \frac{1}{2}. \quad (15)$$

Dividend shocks are zero except at dates 11–14, where they are 2, 4, 6, 8.

Interpretation. A finite sequence of good fundamental shocks starts the run-up. Extrapolation and realization utility then generate the joint dynamics of prices and trading volume.

3.8 Account-level volume regression

The main investor-level empirical specification is

$$\Delta Volume_i = \alpha + \beta_X DOX_i + \beta_D DOD_i + \beta_{XD}(DOX_i \times DOD_i) + \Gamma' Controls_i + \varepsilon_i. \quad (16)$$

- $\Delta Volume_i$ is the ratio of monthly volume at the bubble peak to average monthly volume in the prebubble period.
- DOX_i and DOD_i are measured using 2005–2013 transactions.
- Controls include account size, experience, concentration, volatility seeking, skewness seeking, past returns, demographics, margin status, prior warrant trading, and survey-based characteristics.

Interpretation. This regression asks whether ex ante behavioral traits explain who trades much more during the bubble.

3.9 Turnover and balance decomposition

The paper also replaces $\Delta Volume_i$ with

$$\Delta Turnover_i, \quad \Delta Balance_i, \quad \Delta CrashVolume_i. \quad (17)$$

Interpretation. The decomposition separates the two ways investors can generate more volume: holding more assets or reshuffling assets faster. The crash-volume specification is a placebo-style check.

3.10 Stock-level turnover regression

The stock-level specification is

$$Turnover_{j,t} = \alpha_j + \beta_X DOX_{j,t} + \beta_D DOD_{j,t} + \Gamma' Z_{j,t} + u_{j,t}, \quad (18)$$

where α_j are stock fixed effects and $Z_{j,t}$ includes contemporaneous and lagged returns and turnover controls.

Interpretation. This regression asks whether stocks traded by disposition extrapolators have higher turnover, controlling for permanent differences across stocks.

3.11 Counterfactual estimation of θ and β

The paper estimates the extrapolation and disposition parameters using stock-level retail flows before the bubble:

$$InitialBuys_{i,t} = b_0 \left(b_1 + \frac{\sum_{\tau=1}^T \theta^\tau r_{i,t-\tau}}{\sum_{\tau=1}^T \theta^\tau} \right) + u_{i,t}, \quad (19)$$

$$SubsequentTrades_{i,t} = b_0 \left(b_1 + \frac{\sum_{\tau=1}^T \theta^\tau r_{i,t-\tau}}{\sum_{\tau=1}^T \theta^\tau} - \beta \bar{r}_{i,t} \right) + v_{i,t}. \quad (20)$$

Interpretation. Initial buys identify the extrapolation horizon because they are assumed to be driven mainly by expected returns. Subsequent trades identify the disposition term because current holders trade based on both beliefs and gains/losses.

3.12 Return regressions for price effects

To study prices, the paper estimates predictive and IV-style stock-return regressions such as

$$R_{j,t+1} = \alpha + \beta_X DOX_{j,t} + \Gamma' X_{j,t} + \delta_t + \varepsilon_{j,t}. \quad (21)$$

It also instruments contemporaneous $DOX_{j,t+1}$ using lagged $DOX_{j,t}$.

Interpretation. The goal is to reduce reverse causality: instead of only showing that extrapolators trade stocks whose prices are already rising, the paper asks whether past extrapolator presence predicts future returns.

4 Identification and empirical design (what makes the estimates credible)

4.1 What identifies the volume mechanism

- The central source of variation is cross-investor heterogeneity in behavioral measures estimated *before* the bubble.
- Investors with high DOX and high DOD are compared to investors with high DOX but low DOD, and to all other investors.
- Since the behavioral measures are estimated from 2005–2013, they are not mechanically caused by 2014–2015 bubble trading.

Interpretation. The paper’s key design is predictive rather than contemporaneous: earlier behavioral patterns explain later bubble trading.

4.2 Why the disposition-extrapolator comparison is powerful

- Pure extrapolators and disposition extrapolators both chase winners.
- The difference is that disposition extrapolators are also prone to selling winners.
- Therefore, the comparison isolates the additional volume mechanism created by selling to realize gains.

Interpretation. The mechanism is not just “optimistic investors trade more.” It is specifically that optimism and gain-realization motives interact.

4.3 Controls for alternative investor traits

- Account-level regressions control for account size, experience, concentration, volatility seeking, skewness seeking, and past returns.
- They also control for demographics and survey-based measures of wealth, income, sophistication, investment horizon, and risk tolerance.
- Margin-account and warrant-trading controls address leverage access and prior bubble-like trading experience.

Interpretation. The results are less likely to be driven by simple differences in wealth, sophistication, risk tolerance, leverage, or gambling taste.

4.4 Placebo and crash-period logic

- The model predicts that the interaction of extrapolation and disposition is especially relevant in the run-up, when stocks are generating gains.
- In the crash, gains turn into losses and disposition-prone investors become reluctant to sell, so the same traits should not predict high volume in the same way.
- Table 5’s crash-volume column finds that neither DOX nor DOD significantly explains crash volume.
- The paper also reports that in a quieter market period, the same measures do not explain volume strongly.

Interpretation. The absence of the same relationship outside the run-up supports the idea that the mechanism is bubble-specific.

4.5 Stock-level identification

- Stock-level regressions use weekly variation in which types of investors are buying and selling each stock.
- Stock fixed effects remove persistent stock-level differences.
- Time-clustered standard errors account for common shocks across stocks in the same week.
- Controls for current and lagged returns and turnover reduce concerns that the result simply reflects momentum or mechanically high trading in already-hot stocks.

Interpretation. The stock-level design shows that the mechanism is visible not only across investors, but also across stocks within the bubble.

4.6 Why the price evidence is more suggestive

- A contemporaneous positive relation between extrapolator trading and returns is difficult to interpret, because rising returns may attract extrapolators.
- The paper therefore uses predictive regressions and IV regressions with lagged DOX.
- These designs reduce but do not completely eliminate endogeneity concerns.

Interpretation. The paper is careful: the volume tests are closer to the core identification strategy, while the price tests provide supporting evidence for extrapolative price pressure.

4.7 Remaining limitations

- Investor behavioral types are not randomly assigned.
- The data focus on a large brokerage and regular accounts, not the entire market or all margin/shadow-leverage activity.
- New investors who enter during the bubble are not central to the main design because their prebubble behavior cannot be measured.
- The paper cannot fully separate every possible psychological source of the disposition effect; it uses realization utility as the leading mechanism.

Interpretation. The evidence strongly supports the volume mechanism, but the paper does not claim a fully randomized causal estimate for every component of bubble formation.

5 Tables and figures

5.1 Figure 1: Prices and signals in the baseline model

Read:

- The simulated price initially rises after positive dividend shocks.
- The price continues rising even after fundamentals stop improving because the extrapolative signal remains positive.
- Eventually the value signal becomes sufficiently negative and the bubble reverses.
- The gain relative to the reference price, $P_t - \bar{P}_t$, rises during the run-up, strengthening the disposition motive to sell.

Economic message. Extrapolation explains why prices rise beyond fundamentals; the accumulation of gains explains why holders become willing to sell even while beliefs remain optimistic.

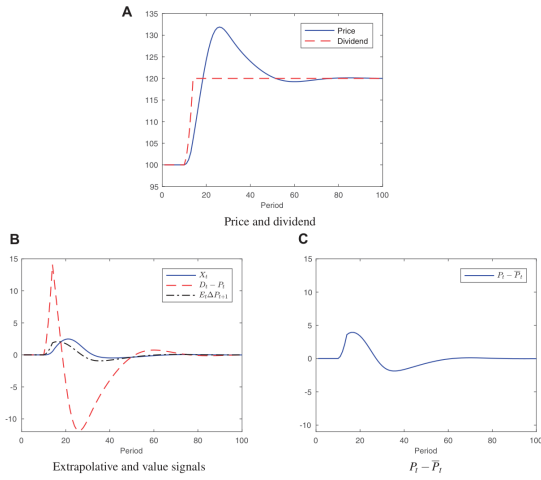


Figure 1
Prices and signals in the baseline case
 In panel A, the dashed line represents the dividend, D_t , and the solid line represents the stock price, P_t . In panel B, the solid line represents X_t ; the dashed line represents $D_t - P_t$; and the dash-dot line represents $E_t \Delta P_{t+1}$, defined as $\gamma X_t + (1 - \gamma)(D_t - P_t)$, where $\gamma = 0.9$. In panel C, the solid line represents the difference between the current stock price and the reference price, $P_t - \bar{P}_t$. There are 101 dates. The dividend shocks are set to zero, except for dates 11 to 14, when the dividend shocks are 2, 4, 6, and 8, respectively. Other parameter values are $\theta = 0.8$, $\beta = 1$, $\sigma_\varepsilon = 2$, $D_0 = 100$, $X_1 = 0$, and $Q = 1/2$.

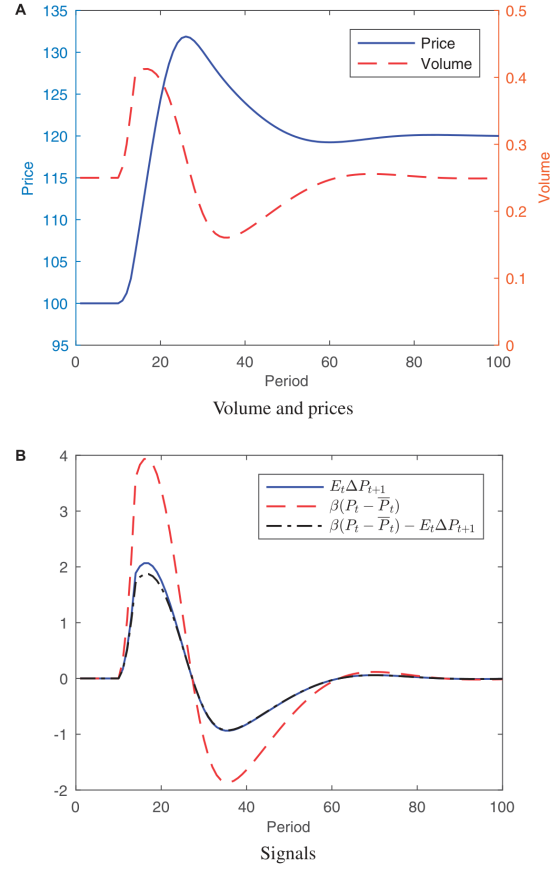


Figure 2
Trading volume in the baseline case
 In panel A, the solid line represents total trading volume, and the dashed line represents the stock price. In panel B, the solid line represents $E_t \Delta P_{t+1}$; the dashed line represents $\beta(P_t - \bar{P}_t)$; and the dash-dot line represents $\beta(P_t - \bar{P}_t) - E_t \Delta P_{t+1}$. There are 101 dates. The dividend shocks are set to zero, except for dates 11 to 14, when the dividend shocks are 2, 4, 6, and 8, respectively. Other parameter values are $\theta = 0.8$, $\beta = 1$, $\sigma_\varepsilon = 2$, $D_0 = 100$, $X_1 = 0$, and $Q = 1/2$.

5.2 Figure 2: Trading volume in the baseline model

Read:

- Volume is hump-shaped and peaks before prices.
- The model has a natural lead-lag pattern: volume peaks when expectations are most optimistic, while price peaks later when optimism has faded to roughly zero.
- Selling pressure comes from the gain-realization term $\beta(P_t - \bar{P}_t)$, not from pessimism.

Economic message. The same rising market creates buyers and sellers at the same time: cash holders chase returns, while stock holders lock in gains.

5.3 Figure 3: Comparative statics

Read:

- Lower θ , meaning stronger sensitivity to recent returns, produces a larger price bubble.
- Higher β , meaning stronger disposition pressure, lowers peak prices but does not eliminate the bubble.
- Higher belief dispersion σ_ε reduces price and volume in this model because demand becomes less responsive to signals.
- Higher γ , meaning more weight on the extrapolative signal, increases peak prices.

Economic message. The model is not a generic disagreement story. Price and volume are driven by extrapolative demand and the disposition-induced willingness to sell, not merely by dispersion in beliefs.

5.4 Figure 4: Prices and trading volume on the Shenzhen Stock Exchange

Read:

- The Shenzhen Component Index rises sharply in the run-up and crashes after mid-June 2015.
- Trading volume rises even more dramatically than prices.
- The figure divides the episode into prebubble, run-up, and crash phases.

Economic message. The empirical setting mirrors the model's object of interest: a bubble with both extreme price movements and extreme trading volume.

5.5 Table 1: Summary statistics of investor characteristics

Read:

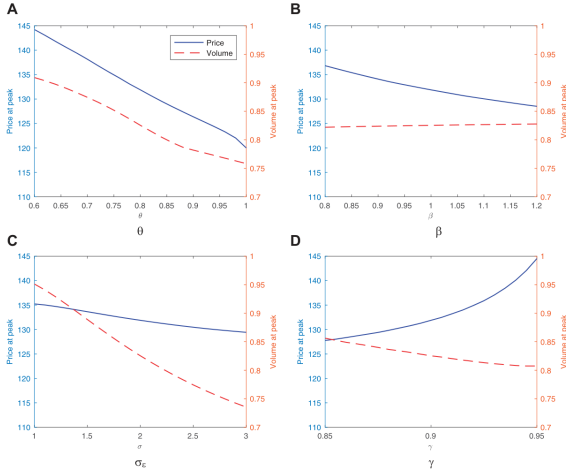


Figure 3
Comparative statics
This figure presents the price and volume at peak under parameters that are different from those of the baseline scenario. There are 101 dates. The dividend shocks are set to zero, except for dates 11 to 14, when the dividend shocks are 2, 4, 6, and 8, respectively. In the baseline scenario, the parameter values are $\theta=0.8$, $\beta=1$, $\sigma_\epsilon=2$, and $\gamma=0.9$. The title of each subfigure is the parameter concerned.

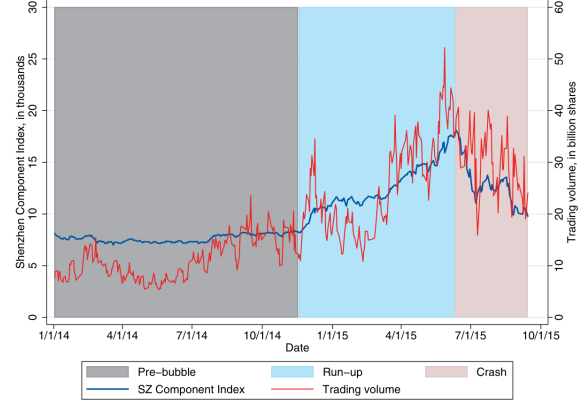


Figure 4
Prices and trading volume at SZSE
The thick line plots the closing price of the Shenzhen Component Index (SZCI; in thousands), and the thin line plots the number of shares traded on the SZSE (in billions; scale on the right axis). The time frame is from January 1, 2014, to September 15, 2015. The shaded areas represent the three stages of the bubble: the prebubble stage, from January 1, 2014, to November 17, 2014; the run-up stage, from November 18, 2014, to June 12, 2015; and the crash stage, from June 13, 2015, to September 15, 2015.

Although the government responded immediately with various measures to prop up the market, the recovery was short-lived; the market plummeted again in mid-August and continued to fall until September.

Given the discussion above, we adopt the following timeline to study this bubble: (1) 2014:01 to 2014:11 is the prebubble period, because price reactions in the market were muted; (2) 2014:12 to 2015:05 is the run-up, manifested by intensive media coverage and strong market reactions; and (3) 2015:06 to 2015:08 is the crash.

- The main sample contains active individual investors with enough pre-2014 trading history to estimate behavior.
- The median account balance is about 0.13 million RMB and the average account balance is about 0.22 million RMB.
- Only around 2% of the sample has a margin account.
- The median investor's average monthly return is negative.

Economic message. The sample is made up mostly of ordinary retail investors rather than large institutional or highly levered accounts.

5.6 Table 2: Summary statistics for extrapolation and disposition measures

Read:

- The DOX measures are positive for most investors, showing widespread buying of recent winners.
- The DOD measures show a strong disposition effect: investors realize gains more readily than losses.
- The correlation between weekly and monthly DOX is high, and the two disposition measures are also related.

Economic message. The two behavioral ingredients in the model are empirically measurable and

Table 1
Summary statistics of sample characteristics

	Min	P5	P25	P50	P75	P95	Max	Mean	SD
AGE	18	28	36	43	51	65	75	44	11
BAL	0.01	0.02	0.06	0.13	0.30	0.72	0.99	0.22	0.22
EXP	1.08	2.83	5.33	7.25	7.92	8.92	8.92	6.63	1.90
COUNT_BUY	14	19	40	85	194	636	3,502	178	289
COUNT_SELL	10	14	32	71	169	574	3,299	157	267
TN	0.0	0.1	0.4	0.8	2.1	8.9	781.4	3.9	31.0
RET	-35.0%	-7.4%	-2.9%	-1.4%	-0.3%	1.7%	18.8%	-1.9%	3.5%
FEMALE	0	0	0	0	1	1	1	0.48	0.50
DUMMY_MARGIN	0	0	0	0	0	0	1	0.02	0.15
DUMMY_CALLS	0	0	0	0	0	1	1	0.15	0.36
DUMMY_PUTS	0	0	0	0	0	1	1	0.11	0.32
DUMMY_A	0	0	0	0	0	0	1	0.02	0.16
DUMMY_B	0	0	0	0	0	1	1	0.13	0.34

This table reports the summary statistics of the main sample of investors. Only accounts opened prior to 2014 are included in the main sample. We also drop accounts that have made fewer than 14 buys or 10 sells. *BAL* is the average RMB holding in millions. *EXP* is the number of years since account open date. *COUNT_BUY* (*COUNT_SELL*) is the number of buys (sells). *TN* is turnover and is calculated by dividing total trading volume by average account balance. *RET* is the average monthly return rate, calculated by dividing total *RMB* return by average *RMB* holding. *DUMMY_MARGIN*, *DUMMY_CALLS*, *DUMMY_PUTS*, *DUMMY_A*, and *DUMMY_B* are dummy variables indicating having a margin account, having traded call warrants, having traded put warrants, having traded A funds, and having traded B funds, respectively. P5, P25, P50, P75, and P95 correspond to the 5th, 25th, 50th, 75th, and 95th percentiles in the distribution.

Table 2
Summary statistics for account characteristics

A. Summary statistics										
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	DOXW	DOXM	DODD	DODR	HHI	VOL	SKEW			
Min	-0.07	-0.11	-0.45	0.33	0.08	0.02	-0.28			
P5	-0.02	-0.02	-0.08	0.81	0.24	0.02	0.00			
P25	0.01	0.04	0.07	1.19	0.43	0.03	0.15			
P50	0.02	0.08	0.16	1.56	0.59	0.03	0.30			
P75	0.04	0.13	0.27	2.18	0.75	0.04	0.56			
P95	0.08	0.23	0.47	4.34	0.93	0.05	1.35			
Max	0.25	0.60	0.81	19.30	1.00	0.20	3.82			
Mean	0.03	0.09	0.17	1.96	0.59	0.03	0.44			
SD	0.03	0.08	0.17	1.52	0.21	0.01	0.47			

B. Correlation matrix										
	DOXW	DOXM	DODD	DODR	HHI	VOL	SKEW	TN	RET	BAL
DOXW										
DOXM	0.78									
DODD	-0.03	-0.02								
DODR	-0.05	-0.02	0.64							
HHI	0.04	0.00	-0.11	-0.33						
VOL	0.20	0.22	-0.08	-0.09	0.07					
SKEW	0.08	0.08	-0.03	-0.04	0.04	0.55				
TN	0.00	-0.02	-0.04	-0.04	0.05	0.03	0.02			
RET	-0.02	0.05	0.09	0.11	-0.05	-0.11	-0.11	-0.09		
BAL	0.00	0.01	-0.10	-0.03	-0.14	0.07	0.04	0.04	0.00	
EXP	0.10	0.21	0.04	0.05	-0.11	0.11	0.00	-0.02	0.12	0.11

DOXW and *DOXM* are degrees of extrapolation based on past weekly returns and monthly returns, respectively, and are calculated as volume-weighted past returns based on all initial buys. *DODD* and *DODR* are degrees of disposition based on the difference and ratio, respectively, between *PGR* and *PLR*, where *PGR* (Proportion of Gains Realized) is calculated by dividing the number of realized winners by the total number of winners on days of sales and *PLR* (Proportion of Losses Realized) is similarly calculated. *HHI* is the Herfindahl-Hirschman index based on monthly holdings. *VOL* is calculated as volume-weighted past volatility. *SKEW* is calculated as volume-weighted past skewness. *TN* is turnover and is calculated by dividing total trading volume by average account balance. *RET* is the average monthly return rate, calculated by dividing total *RMB* return by average *RMB* holding. *BAL* is the average RMB holding in millions. *EXP* is the number of years since account open date. All variables are constructed based on transactions from 2005 to 2013. P5, P25, P50, P75, and P95 correspond to the 5th, 25th, 50th, 75th, and 95th percentiles in the distribution.

prevalent before the bubble begins.

5.7 Table 3: Extrapolation and disposition across investor groups

Read:

- Extrapolation and disposition vary across age, education, and gender groups.
- The disposition effect is stronger among older investors and women in these descriptive averages.
- Extrapolation differences are present but relatively modest across demographic groups.

Economic message. Demographics matter enough that they should be controlled for, but the behavioral measures are not simply demographic labels.

Table 3
Extrapolation and disposition effect across investor groups

	DOXW	DOXM	DODD	DODR	Obs.
<i>A. Age</i>					
30 or younger	0.027	0.078	0.162	1.804	19,612
30-39	0.026	0.081	0.170	1.891	78,485
40-49	0.026	0.085	0.172	1.945	85,165
50-59	0.027	0.091	0.176	2.056	51,940
60-69	0.027	0.093	0.161	2.040	24,514
70 or older	0.029	0.097	0.154	2.008	5,485
<i>B. Education</i>					
Doctoral	0.028	0.093	0.183	2.002	6,521
Master's	0.025	0.079	0.152	1.891	5,395
Bachelor's	0.027	0.086	0.164	1.909	75,969
3-year college	0.027	0.087	0.175	1.981	83,793
Professional school	0.026	0.084	0.174	1.977	21,841
High school	0.026	0.086	0.173	1.953	46,357
Middle school	0.026	0.086	0.170	1.955	25,469
Others	0.025	0.083	0.177	2.008	10,760
<i>C. Gender</i>					
Male	0.027	0.085	0.161	1.832	303,530
Female	0.028	0.093	0.187	2.100	280,329

This table reports the average degrees of extrapolation and disposition across demographic groups. *DOXW* and *DOXM* are degrees of extrapolation based on past weekly returns and monthly returns, respectively, and are calculated as volume-weighted past returns based on all initial buys. *DODD* and *DODR* are degrees of disposition based on the difference and ratio, respectively, between *PGR* and *PLR*, where *PGR* (Proportion of Gains Realized) is calculated by dividing the number of realized winners by the total number of winners on days of sales and *PLR* (Proportion of Losses Realized) is similarly calculated. All variables are constructed based on transactions from 2005 to 2013.

Table 4
Distribution of ownership breadth in the cross-section of individual stocks, 2014-2015

Date	Min	P5	P25	P50	P75	P95	Max
Jan. 2014	0.01%	0.02%	0.04%	0.08%	0.14%	0.41%	2.50%
Feb. 2014	0.01%	0.02%	0.04%	0.08%	0.14%	0.41%	2.48%
Mar. 2014	0.01%	0.02%	0.04%	0.08%	0.14%	0.42%	2.47%
Apr. 2014	0.01%	0.02%	0.04%	0.08%	0.14%	0.41%	2.47%
May 2014	0.01%	0.02%	0.04%	0.08%	0.14%	0.41%	2.46%
Jun. 2014	0.01%	0.02%	0.04%	0.08%	0.14%	0.41%	2.45%
Jul. 2014	0.01%	0.02%	0.05%	0.08%	0.14%	0.41%	2.44%
Aug. 2014	0.01%	0.02%	0.05%	0.08%	0.14%	0.41%	2.43%
Sep. 2014	0.01%	0.02%	0.05%	0.08%	0.14%	0.41%	2.41%
Oct. 2014	0.01%	0.02%	0.04%	0.08%	0.14%	0.41%	2.40%
Nov. 2014	0.01%	0.02%	0.04%	0.08%	0.14%	0.41%	2.37%
Dec. 2014	0.01%	0.02%	0.04%	0.07%	0.14%	0.41%	2.26%
Jan. 2015	0.01%	0.02%	0.04%	0.07%	0.14%	0.41%	2.73%
Feb. 2015	0.01%	0.02%	0.04%	0.07%	0.14%	0.41%	2.92%
Mar. 2015	0.01%	0.02%	0.04%	0.07%	0.13%	0.42%	2.87%
Apr. 2015	0.01%	0.02%	0.04%	0.07%	0.13%	0.42%	2.78%
May 2015	0.01%	0.02%	0.04%	0.07%	0.13%	0.41%	2.81%
Jun. 2015	0.01%	0.02%	0.04%	0.07%	0.12%	0.40%	3.26%
Jul. 2015	0.01%	0.02%	0.04%	0.07%	0.13%	0.39%	2.94%
Aug. 2015	0.01%	0.02%	0.04%	0.07%	0.13%	0.38%	2.76%
Sep. 2015	0.01%	0.02%	0.04%	0.07%	0.13%	0.37%	2.73%
Oct. 2015	0.01%	0.02%	0.04%	0.07%	0.13%	0.38%	2.58%
Nov. 2015	0.01%	0.02%	0.04%	0.07%	0.13%	0.38%	2.44%
Dec. 2015	0.01%	0.02%	0.04%	0.07%	0.13%	0.38%	2.40%

In each month in 2014-2015, we calculate each stock's ownership breadth, defined by dividing the number of investors holding that stock by the number of investors in the population.

5.8 Table 4: Ownership breadth and heterogeneous holdings

Read:

- Ownership breadth is very low across stocks.
- The median stock is held by only about 0.07%–0.08% of investors.
- Even the 95th percentile of ownership breadth is typically around 0.38%–0.42%.

Economic message. Investors start with different portfolios. This heterogeneity is necessary for the model because some investors must be current holders with gains while others are cash holders or nonholders chasing returns.

5.9 Figure 5: Evolution of volume by investor group

Read:

- Disposition extrapolators increase volume the most during the run-up.
- At the peak, their monthly volume rises by almost 800% from the prebubble baseline.
- Pure extrapolators increase holdings aggressively but do not rebalance as fast.
- The decomposition shows that extrapolation mainly increases balances, while disposition mainly increases turnover.

Economic message. The figure visually confirms the paper’s mechanism: high volume requires both buying pressure and rapid gain realization.

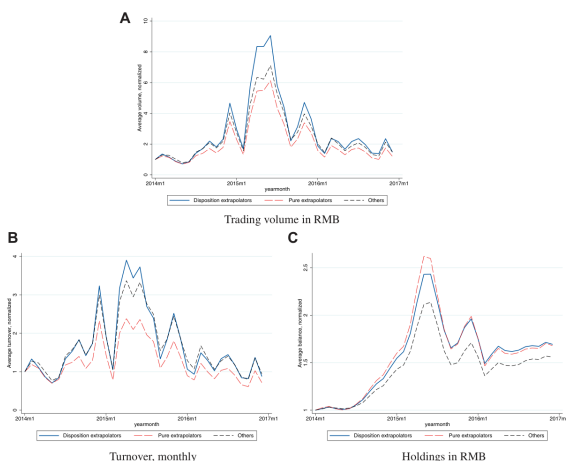


Figure 5
Evolution of volume by group

The three lines represent the evolution of volume for three investor groups: disposition extrapolators, pure extrapolators, and other investors. Disposition extrapolators have both *DOX* and *DOD* above the median; pure extrapolators have *DOX* above the median and *DOD* below; and the rest are classified as other investors. For all groups, volume, turnover, and balance are normalized to one at the beginning of 2014.

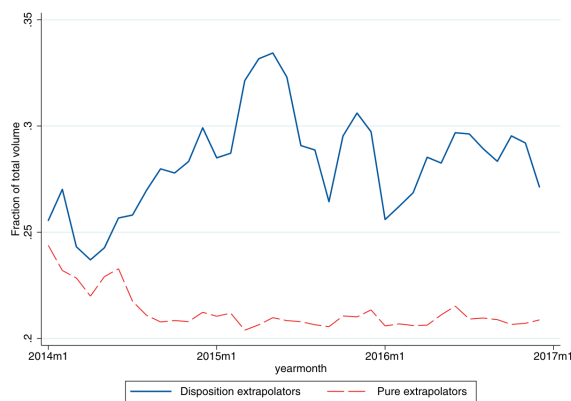


Figure 6
Decomposition of total volume by group

This plots the composition of total volume. The solid line represents the fraction of volume from disposition extrapolators, and the dashed line represents the fraction from pure extrapolators. Disposition extrapolators have both *DOX* and *DOD* above the median and pure extrapolators have *DOX* above the median and *DOD* below.

5.10 Figure 6: Decomposition of total volume by group

Read:

- Disposition extrapolators account for about 25% of total volume before the bubble.
- Their share rises to about 34% at the bubble peak.
- Pure extrapolators' share falls from about 25% to roughly 20%.

Economic message. The bubble's volume is not just a marketwide rise in activity; the composition of volume shifts toward investors who combine extrapolation with the disposition effect.

5.11 Table 5: Account-level volume regressions

Read:

- In the baseline regression, a one-standard-deviation increase in DOX predicts a 4.02 increase in relative peak volume; a one-standard-deviation increase in DOD predicts a 4.60 increase.
- In the full specification, DOX, DOD, and their interaction remain statistically significant.
- The turnover/balance decomposition shows that DOX mainly predicts balance growth, while DOD mainly predicts turnover growth.
- The crash-volume column does not reproduce the run-up relationship.

Economic message. Investor-level regressions confirm the group-level pattern and show that the mechanism survives a rich set of controls.

5.12 Table 6: Stock-level turnover regressions

Read:

- Stocks bought more by extrapolators and sold more by disposition-prone investors have higher turnover.
- Baseline coefficients are 0.04 for $DOX_{j,t}$ and 0.02 for $DOD_{j,t}$.
- Both remain positive and significant after adding return and turnover controls.
- Stock fixed effects absorb permanent differences across stocks.

Economic message. The mechanism operates not only across investor types but also across stocks: stocks exposed to disposition extrapolators trade more.

5.13 Table 7: Model counterfactuals

Read:

- Estimated parameters are $\theta = 0.64$ and $\beta = 0.54$ in the full sample.

Table 5
Explaining account-level trading volume using extrapolation and the disposition effect

	ΔVolume_i			$\Delta \text{Turnover}_i$		$\Delta \text{Balance}_i$	$\Delta \text{CrashVolume}_i$
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
DOX_i	4.02*** (10.31)	3.69*** (9.56)	3.64*** (9.44)	2.64*** (5.56)	-0.02 (-0.10)	0.32*** (17.33)	-0.075 (-0.683)
DOD_i	4.60*** (13.31)	4.32*** (12.27)	4.14*** (11.81)	3.65*** (7.84)	1.96*** (11.24)	-0.05*** (-4.04)	0.012 (0.124)
$\text{DOX}_i * \text{DOD}_i$	0.84*** (2.94)	0.72*** (2.63)	0.71*** (2.59)	0.76** (2.15)	0.27** (1.99)	-0.04*** (-4.61)	-0.105* (-1.664)
BAL_i	-19.60*** (-22.44)	-18.77*** (-21.08)	-14.96*** (-13.61)	-0.60 (-1.45)	-1.39*** (-32.24)	-3.68*** (-13.261)	-3.68*** (-13.261)
EXP_i	2.69*** (31.98)	2.84*** (32.83)	3.25*** (30.55)	1.33*** (34.34)	0.04*** (9.14)	0.690*** (12.303)	0.690*** (12.303)
HHI_i	0.80 (0.75)	-0.18 (-0.17)	2.70** (2.08)	-3.67*** (-7.74)	1.03*** (20.71)	4.352*** (13.277)	4.352*** (13.277)
VOL_i	-122.23*** (-7.35)	-118.97*** (-7.16)	-80.00*** (-3.91)	-69.62*** (-10.10)	6.15*** (7.09)	-1.888 (-0.326)	-1.888 (-0.326)
SKEW_i	1.20** (2.20)	1.31** (2.42)	1.14* (1.70)	0.63*** (2.96)	-0.02 (-0.56)	0.620*** (3.431)	0.620*** (3.431)
RET_i	-13.35*** (-3.22)	-12.85*** (-3.10)	4.75 (1.11)	6.69*** (4.45)	-2.18*** (-7.07)	-5.487*** (-3.577)	-5.487*** (-3.577)
Other controls							
Demographics	No	No	Yes	Yes	Yes	Yes	Yes
Margin account, dummy	No	No	No	Yes	Yes	Yes	Yes
Traded warrants before, dummy	No	No	No	Yes	Yes	Yes	Yes
Survey-based characteristics	No	No	No	Yes	Yes	Yes	Yes
Constant	26.59*** (55.20)	14.81*** (12.71)	12.52*** (10.31)	3.34 (1.14)	4.70*** (4.53)	1.52*** (11.79)	1.453* (1.800)
N	439,853	439,798	439,798	252,907	252,907	215,146	215,146
R^2	.003	.005	.006	.010	.013	.016	.007

This table reports the results from regressing changes in trading volume, turnover, and balance on degrees of extrapolation and disposition. DOX is the degree of extrapolation, calculated as volume-weighted past monthly returns based on all initial buys. DOD is the degree of disposition, calculated as the ratio of PGR to PLR, where PGR (Proportion of Gains Realized) is calculated by dividing the number of realized winners by the total number of winners on days of sales and PLR (Proportion of Losses Realized) is similarly calculated. BAL is the average RMB holding in millions. EXP is the number of years since account open date. HHI is the Herfindahl-Hirschman index based on monthly holdings. VOL is calculated as volume-weighted past volatility. SKEW is calculated as volume-weighted past skewness. RET is the average monthly return rate, calculated by dividing total RMB return by average RMB holding. DOX to RET are constructed based on transactions from 2005 to 2013. Demographic variables include gender, age, and education. Survey-based characteristics include answers to questions related to expected returns and risks; self-reported wealth, income, and sophistication; investment horizon, experience, and objectives; and short-term and long-term tolerances for losses. ΔVolume is calculated as the ratio of monthly volume at peak (2015:05) to the average monthly volume in the prebubble period from 2014:01 to 2014:11. $\Delta \text{Turnover}$ and $\Delta \text{Balance}$ are similarly calculated. $\Delta \text{CrashVolume}$ is calculated as the ratio of monthly volume at trough (2015:09) to the average monthly volume in the prebubble period from 2014:01 to 2014:11. Robust t -statistics appear in parentheses. *** $p < .01$; ** $p < .05$; * $p < .1$.

Table 6
Explaining stock-level turnover using extrapolation and the disposition effect

	Turnover (t)			
	(1)	(2)	(3)	(4)
$\text{DOX}(t)$	0.04*** (14.30)	0.04*** (9.34)	0.01*** (2.89)	0.01*** (2.92)
$\text{DOD}(t)$	0.02*** (7.76)	0.01*** (6.32)	0.01*** (5.13)	0.01*** (5.53)
Return (t)		0.28*** (3.97)	0.38*** (6.44)	0.40*** (7.31)
Return ($t-1$)			0.38*** (10.09)	0.25*** (6.70)
Return ($t-2$)			0.28*** (6.54)	0.19** (2.37)
Return ($t-3$)			0.18*** (4.37)	0.00 (0.10)
Return ($t-4$)			0.12*** (2.86)	0.02 (0.44)
Turnover ($t-1$)				0.37*** (7.76)
Turnover ($t-2$)				0.09*** (4.84)
Turnover ($t-3$)				0.05 (1.48)
Turnover ($t-4$)				-0.05 (-1.05)
Return ($t-5$) to ($t-12$)	No	No	Yes	Yes
Turnover ($t-5$) to ($t-12$)	No	No	Yes	Yes
Stock FE	Yes	Yes	Yes	Yes
Time-clustered SE	Yes	Yes	Yes	Yes
N	63,639	63,639	63,307	63,307
R^2	.50	.52	.62	.70

This table reports panel regression results by regressing weekly stock-level turnover on weekly stock-level measures of extrapolation and disposition. A stock's turnover in a given week is calculated by dividing the total RMB trading amount by its market capitalization. Stock-level degree of extrapolation is calculated as the buy-volume-weighted average degree of extrapolation in a given week and stock-level degree of disposition is calculated as the sell-volume-weighted average degree of disposition in a given week. The sample period is from 2014:12 to 2015:05. Robust t -statistics appear in parentheses. *** $p < .01$; ** $p < .05$; * $p < .1$.

- With all investors, model-implied peak price is 190 and peak volume is 0.46.
- With no extrapolation and no disposition bias, peak price is 120 and peak volume is 0.25.
- Removing disposition extrapolators gives peak price 131 and peak volume 0.36.

Economic message. The proposed mechanism is quantitatively important. Relative to the no-disposition-extrapolator benchmark, it raises peak volume by roughly 30%.

5.14 Table 8: Stock returns and extrapolator exposure

Read:

- Contemporaneous regressions are positive but hard to interpret because of reverse causality.
- Predictive regressions show that higher current DOX predicts higher next-week returns during the run-up.
- The IV specification also finds a positive run-up effect.
- During the crash, predictive and IV estimates become negative.

Economic message. The price evidence supports the extrapolative-bubble interpretation: extrapolators appear to push prices up in the run-up and down in the crash, although the paper is cautious about full causality.

Table 7
Model counterfactuals

	All investors (1)	No bias (2)	Benchmark No disposition extrapolators (3)
Empirical estimates			
θ	0.64 (0.003)	0	0.92 (0.001)
β	0.54 (0.002)	0	0.41 (0.001)
Model outputs			
Peak price	190	120	131
Peak volume	0.46	0.25	0.36

This table reports peak price and volume under three sets of parameters. θ represents the degree of extrapolation and β represents the degree of disposition. In column 1, using the full sample of investors including disposition extrapolators, we estimate θ and β from the following two equations:

Table 8
Regressing stock returns on stock-level measures of extrapolation and disposition

	A. Return ($t+1$), run-up (%)				B. Return ($t+1$), crash (%)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\overline{DOX}(t+1)$	2.90*** (9.38)			0.70** (2.07)	3.69*** (4.91)			-4.49*** (-3.20)
$\overline{DOX}(t)$		0.35** (2.26)	0.35** (2.25)			-1.83** (-3.02)	-1.86*** (-3.13)	
Return (t)	-0.13** (-2.18)	-0.09 (-1.42)	-0.09 (-1.41)	-0.10 (-1.57)	0.03 (0.16)	0.05 (0.26)	0.05 (0.27)	0.06 (0.33)
BETA (t)	-0.10 (-0.37)	-0.33 (-1.08)	-0.11 (-0.37)	-0.15 (-0.53)	0.21 (0.28)	-0.94 (-1.25)	-0.75 (-0.80)	-0.95 (-0.97)
SIZE (t)	-0.00 (-1.65)	-0.01*** (-3.32)			0.01 (1.07)	0.00 (0.06)		
B/M (t)	0.14 (1.59)	-0.05 (-0.60)			0.47** (2.41)	0.09 (0.47)		
Turnover (t)	-3.91** (-2.19)	-0.92 (-0.49)	-1.50 (-0.74)	-1.29 (-0.64)	-10.64 (-1.50)	-5.07 (-0.66)	-5.46 (-0.65)	-4.74 (-0.55)
FLOAT (t)	0.00 (0.66)	0.00 (1.29)	-0.00 (-0.85)	-0.00 (-0.71)	0.00 (0.51)	0.00 (0.75)	0.00 (0.53)	0.00 (0.18)
VOL (t)	-0.00 (-0.46)	-0.00 (-0.27)	0.00 (0.42)	0.00 (0.32)	-0.00 (-0.42)	-0.00 (-0.49)	-0.00 (-0.63)	-0.00 (-0.32)
Board FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Size and B/M bins	No	No	Yes	Yes	No	No	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time-clustered SE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	59,287	59,277	59,277	59,062	22,939	22,944	22,944	22,785
R ²	.13	.04	.04	.08	.08	.05	.05	.10

This table reports panel regression results by regressing weekly future returns on weekly stock-level exposure to extrapolation. Stock-level exposure to extrapolation is calculated as the buy-volume-weighted average degree of extrapolation in a given week. BETA is the market beta. SIZE is the market capitalization in RMB. B/M is the ratio of book value to market value. Turnover is calculated by dividing trading amount by total market capitalization. FLOAT is the number of tradable shares. VOL is the number of shares traded. Robust t -statistics appear in parentheses. *** $p < .01$; ** $p < .05$; * $p < .1$.

6 Short summary

- The paper explains high bubble volume by combining extrapolative beliefs with the disposition effect.
- Extrapolation makes investors buy stocks with strong past returns.
- The disposition effect makes investors sell those stocks after continued good performance creates gains.
- This interaction creates rapid portfolio cycling, not simply buy-and-hold optimism.
- The empirical design measures investor behavior before the bubble and tests whether those traits predict trading during the bubble.
- Disposition extrapolators trade much more than pure extrapolators during the run-up.
- Investor-level and stock-level regressions support the same mechanism.
- Counterfactual model estimates suggest the mechanism increases peak volume by about 30%.
- Price regressions provide evidence consistent with extrapolators driving run-up returns and crash declines.

7 Conclusion

7.1 Core conclusion

Liao, Peng, and Zhu show that bubble volume is not merely a by-product of optimism. In their framework, extrapolative beliefs create demand for rising stocks, while the disposition effect creates

supply from investors who want to realize gains. The interaction between the two produces the very high turnover observed in bubbles.

7.2 Economic intuition

An extrapolator asks: “Did this stock go up recently?” A disposition-prone investor asks: “Can I lock in a gain?” A disposition extrapolator asks both questions. During a run-up, the answer to both is often yes, so the same investor type repeatedly buys recent winners and sells after gains. This generates a trading frenzy.

7.3 Takeaway

The paper’s main contribution is to connect a behavioral model to granular trading data. It shows that the combination of nonstandard beliefs and nonstandard preferences can explain both sides of a bubble: the price pressure that pushes assets above fundamentals and the excessive volume that makes bubbles costly for households.